

CURRICULUM VITAE

1. PERSONAL DATA

Name: **SAJID UR REHMAN**

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2. ACADEMIC/PROFESSIONAL PARTICULARS

(a) Field of Specialization: Materials Science, Energy Conversion and Storage Materials and Computational Codes Expert (VASP, CASTEP).

(b) Academic Qualifications

- ✓ Postdoctoral Fellowship, Minzu University of China, Beijing, China
- ✓ Ph.D., Institute of Semiconductor, University of Chinese Academy of Sciences (UCAS), Beijing, China.
- ✓ M.Phil. (Physics), University of Lahore, Pakistan.
- ✓ MS (Computer Science), Government College University Lahore, Pakistan
- ✓ B.Sc. (Hons.) Computational Physics, University of the Punjab, Pakistan.

(c) Academic Awards and Achievements

- ✓ Winner of CAS-TWAS scholarship in PhD at the University of Chinese Academy of Sciences (UCAS), China (2016-2020).
- ✓ Excellent International Student Awards in Ph.D. at the University of Chinese Academy of Sciences (UCAS), China, 2018.
- ✓ Excellent International Student Awards in Ph.D. at the University of Chinese Academy of Sciences (UCAS), China, 2019.
- ✓ Excellent International Graduate Awards in Ph.D. at the University of Chinese Academy of Sciences (UCAS), China, 2020.
- ✓ Chinese Government Scholarship for Outstanding International Students, awarded by the Ministry of Education of China, 2020.
- ✓ Principal Investigator, Beijing Natural Science Foundation (Grant No. IS24029), 132K RMB (2024-2026).

(d) Language Proficiency

- ✓ English (Advanced level proficiency in writing/spoken/reading)
- ✓ Urdu (Advanced level proficiency in writing/spoken/reading)
- ✓ Chinese (Elementary in listening and speaking)
- ✓ Arabic (reading proficiency)

3. CAREER DETAILS

Academic Positions Held

- ✓ Post-Doc Fellowship, School of Science, Minzu University of China, *10/2020 – todate.*
- ✓ Lecturer, Department of Physics, The University of Lahore, Pakistan, *08/2016 – 12/2016.*
- ✓ Senior Science Teacher (Physics), Government High School Baghbanpura Lahore, *04/2012 – 08/2016*
- ✓ Research Assistant, Center for High Energy Physics, University of the Punjab, Pakistan, *03/2008-07/2008*

4. RESEARCH

(a) Research Interests

- ✓ Energy Storage and Conversion Devices
- ✓ Electronic and Optical properties.
- ✓ Photocatalysis & Electrocatalysis
- ✓ Thermoelectric Properties.
- ✓ Metal Ion Batteries.

5. Skills & Expertise

(a) Computational

- ✓ VASP/CASTEP
- ✓ Phonopy/phono3py
- ✓ HiPhive/MACE
- ✓ Boltztrap2/AMSET
- ✓ Python

(b) Experimental

- ✓ Hydrothermal & Microwave Synthesis
- ✓ Characterization (XRD, SEM, TEM etc)
- ✓ Electrochemical workstation
- ✓ LAND Battery testing system
- ✓ Glove Box

6. List of Publications *(in chronological order, starting with the most recent)*

(a) Selected Published Articles

1. Haris Habib, Houda El Hyani, Sajid Ur Rehman, Hamid Ullah, Ke-Fan Wang. " Predicting long-term stability of O₂ and H₂O adsorption on the perovskite Cs₂SnI₆ surfaces: A first-principles approach" **Surfaces and Interfaces** (2026).
2. Zhanyao Xu, Sajid Ur Rehman, Haris Habib, Zeeshan Tariq, Hafiz Muhammad Naeem Ullah, Xiaoming Zhang, Bakhtiar Ul Had, Yang Wang, Chuanbo Li, Honglian Guo. " 2D pentagonal MgTe₂ (α , β , γ) monolayers as single atom catalyst for efficient CO₂ reduction" **Surfaces and Interfaces** (2026). (Equal Contribution)
3. Zhanyao Xu, Sajid Ur Rehman, Yuan Xu, Haris Habib, Zeeshan Tariq, Hafiz Muhammad Naeem Ullah, Yang Wang, Chuanbo Li, and Xiaoming Zhang. "Novel pentagonal MgX₂ (X= O, S, Se, Te) monolayers: promising photocatalysts for overall water splitting and CO₂ reduction." **Inorganic Chemistry Frontiers** (2026). (Equal Contribution)
4. Ali Tariq, Zeeshan Tariq, Hafiz Muhammad Naeem Ullah, Sajid Ur Rehman, Xiaoming Zhang, Jun Zheng, and Yuhua Zuo. "Carbon-Coated CoV₂O₆·2H₂O as a High-Capacity Anode for Lithium-Ion Batteries." *Colloids and Surfaces A: Physicochemical and Engineering Aspects* (2026): 140238.
5. Sajid Ur Rehman, Haris Habib, Zeeshan Tariq, Hafiz Muhammad Naeem Ullah, Faheem K. Butt, Xiaoming Zhang, Chuanbo Li, "Novel Two-dimensional Tellurophosphates: Visible Light Photocatalysts for Water Splitting and Carbon Dioxide Reduction", **Surfaces and Interfaces** (2024):105255.
6. Zia Ur Rehman, Sajid Ur Rehman, Muhammad Bilal, Faheem K. Butt, Asif Hussain, Jawad Ahmad Jrar, Kewang Zheng et al. "Engineering of Interfacial Electric Field by g-C₃N₄/ZnSnO₃ Heterojunction for Excellent Photocatalytic Applications." *Journal of Cleaner Production* (2024): 143258.
7. Zia Ur Rehman, Muhammad Bilal, Faheem K. Butt, Sajid Ur Rehman, Zeeshan Asghar, Kewang Zheng, Yongcai Zhang, Xiaoyong Xu, Jianhua Hou, and Xiaozhi Wang. "Selenium doping to improve internal electric field for excellent photocatalytic efficiency of g-C₃N₄ nanosheets." **Separation and Purification Technology** 350 (2024): 128001
8. Haris Habib, Houda El Hyani, Sajid Ur Rehman, Ke-Fan Wang, "Two-dimensional GaN-SnX/X₂ (X=S, Se) heterostructures: A promising visible light driven photocatalysts for overall water splitting." *International Journal of Hydrogen Energy* 191 (2025), 152264. (Corresponding Author)

9. Zhenghua Li, **Sajid ur Rehman**, Syeda Sitwat Batool, Hui Zhou, Yang Wang, Chuanbo Li, and Xiaoming Zhang. "High-performance Ethanol Detection Achieved by WO₃/Co₃O₄ Composite Heterojunctions with Synergistic pn Junction Features." *Sensors and Actuators B: Chemical* (2025): 137929. (**Equal Contribution**)
10. Haris Habib, **Sajid Ur Rehman**, Houda El Hyani, Muhammad Nawaz Sharif, Furui Tan, and Ke-Fan Wang. "Degradation Pathways in Perovskite Solar Cells: Strategies for Enhancing Stability." ***Energy Technology***: 2500137.
11. Zia Ur Rehman, **Sajid Ur Rehman**, Faheem K. Butt, Yongcai Zhang, Jianhua Hou, and Xiaozhi Wang. "Nitrogen-defect containing Si-C/g-C₃N₄ heterostructure with improved charge transfer kinetics for photocatalytic H₂ production and CO₂ reduction." ***Renewable Energy*** (2025): 124536.
12. Zia Ur Rehman, Muhammad Bilal, **Sajid Ur Rehman**, Faheem K. Butt, Zeeshan Asghar, Yongcai Zhang, Xiaoyong Xu, Jianhua Hou, and Xiaozhi Wang. "Boron doped g-C₃N₄ porous nanosheets to increase electron-hole pair generation for excellent photocatalytic H₂ production and CO₂ reduction." ***Separation and Purification Technology*** 354 (2025): 129535.
13. Xiao Liu, Ming Huang, Xiongfeng Zou, Wajid Ali, **Sajid Ur Rehman**, Juan Li, Ziwei Li, Li Xiang, and Anlian Pan. "Alcohol-sensitive MoS₂ optoelectronic synapses for mimicking human-like visual adaptation." ***InfoMat*** (2025): e70019.
14. Hafiz Muhammad Naeem Ullah, Nouraz Mushtaq, **Sajid Ur Rehman**, Zeeshan Tariq, S. S. Ali, Muhammad Tahir, Chuanbo Li, Xiaoming Zhang, and Junbai Li. "Manganese doped tailored cobalt sulfide as an accelerated catalyst for oxygen evolution reaction." ***Journal of Colloid and Interface Science*** 678 (2025): 1087-1095.
15. Wajid Ali, Liuli Yang, Yunfei Xie, Hao Song, Ming Huang, **Sajid Ur Rehman**, Ziwei Li, Zahir Muhammad, and Anlian Pan. "Magnetic Proximity Induced Giant Enhancement of Valley Polarization and Zeeman Splitting in WS₂/Fe₃GaTe₂ Heterostructures." ***Nano Letters*** (2025).
16. Syeda Sitwat Batool, Muhammad Umair Hassan, Peter G. Oduor, Syeda Maria Batool, **Sajid-Ur-Rehman**, Xiaoming Zhang, and Chuanbo Li. "Graphene/TiO₂ Nanofiber Hybrids for Multispectral Photodetectors." ***ACS Applied Nano Materials*** 8, no. 29 (2025): 14677-14688.
17. **Sajid Ur Rehman**, Haris Habib, Zeeshan Tariq, Hafiz Muhammad Naeem Ullah, Faheem K. Butt, Xiaoming Zhang, Chuanbo Li, "Novel Two-dimensional Tellurophosphates: Visible Light Photocatalysts for Water Splitting and Carbon Dioxide Reduction", ***Surfaces and Interfaces*** (2024):105255.
18. Zia Ur Rehman, **Sajid Ur Rehman**, Muhammad Bilal, Faheem K. Butt, Asif Hussain, Jawad Ahmad Jrar, Kewang Zheng et al. "Engineering of Interfacial Electric Field by g-C₃N₄/ZnSnO₃ Heterojunction for Excellent Photocatalytic Applications." ***Journal of Cleaner Production*** (2024): 143258.

19. Jingjing Wang, Sajid Ur Rehman, Zeeshan Tariq, Bin Zou, Xiaoming Zhang, Faheem K. Butt, and Chuanbo Li. "Two-dimensional CrSe₂/GaN heterostructures for visible-light photocatalysis with high utilization of solar energy." **International Journal of Hydrogen Energy** 51 (2024): 382-395. (Equal Contribution)
20. Zeeshan Tariq, Haris Habib, Sajid Ur Rehman, Hafiz Muhammad Naeem Ullah, Ali Tariq, Faheem K. Butt, Chuanbo Li, and Xiaoming Zhang. "Promising two dimensional XSe₂/SnS (X= Pt, Zr, Hf) vdW heterostructures for overall water splitting with higher carrier mobilities." **International Journal of Hydrogen Energy** 87 (2024): 669-677.
21. Sadaf Noureen, Sajid Ur Rehman, Syeda Maria Batool, Junaid Ali, Qifeng Zhang, Syeda Sitwat Batool, Yang Wang, and Chuanbo Li. "Tailoring Bi₂Se₃ Topological Insulator for Visible-NIR Photodetectors with Schottky Contacts Using Liquid Phase Exfoliation." **ACS Applied Materials & Interfaces** 16, 6 (2024), 8158–8168.
22. Zia Ur Rehman, Muhammad Bilal, Faheem K. Butt, Sajid Ur Rehman, Zeeshan Asghar, Kewang Zheng, Yongcai Zhang, Xiaoyong Xu, Jianhua Hou, and Xiaozhi Wang. "Selenium doping to improve internal electric field for excellent photocatalytic efficiency of g-C₃N₄ nanosheets." **Separation and Purification Technology** 350 (2024): 128001
23. Zeeshan Tariq, Sajid Ur Rehman, Faheem K. Butt, Xiaoming Zhang, and Chuanbo Li. "High carrier mobility ZrSSe/SnX (X= S, Se, S₂, Se₂) vdW heterostructures for photocatalytic overall water splitting: A first-principles study." **Molecular Catalysis** 553 (2024): 113716.
24. Sajid Ur Rehman, Zeeshan Tariq, Bin Zou, Faheem K. Butt, Xiaoming Zhang, Shuai Feng, Bakhtiar Ul Haq, and Chuanbo Li. "Type-I/Type-II Transition of MoSe₂/g-GaN van der Waals heterostructures mediated by biaxial strain and electric field for overall water splitting." **Materials Science and Engineering: B** 288 (2023): 116195.
25. Sajid Ur Rehman, Zeeshan Tariq, Faheem K. Butt, Xiaoming Zhang, Bakhtiar Ul Haq, and Chuanbo Li. "Novel two-dimensional MC₂N₄ (M= Cr, Mo, W) monolayers for overall water splitting with high visible-light absorption." **Solar Energy** 241 (2022): 416-427.
26. Jing Wang, Sajid Ur Rehman, Yang Xu, Binzhou Zuo, Haohang Cheng, Lingshan Guo, Bin Zou, Xiaoming Zhang, and Chuanbo Li. "Two-dimensional antimony selenide (Sb₂Se₃) nanosheets prepared by hydrothermal method for visible-light photodetectors." **Solar Energy** 233 (2022): 213-220. (Equal Contribution)
27. Bakhtiar Ul Haq, Salem AlFaify, R. Ahmed, Faheem K. Butt, Muhammad Tahir, Sajid Ur Rehman, M. M. Alsardia, and Se-Hun Kim. "Physical properties of novel Tin-chalcogenides heterostructures: A first-principles study." **Materials Science in Semiconductor Processing** 149 (2022): 106820.
28. Jingjing Wang, Sajid Ur Rehman, Zeeshan Tariq, Xiaoming Zhang, Jun Zheng, Faheem K. Butt, Chuanbo Li, "Pristine and Janus Chromium Dichalcogenides: Potential Photocatalysts for

Overall Water Splitting in Wide Solar Spectrum Under Strain and Electric Field.” **Solar Energy and Solar Cell** 230 (2021): 111258. (Equal Contribution)

29. Zeeshan Tariq, Sajid Ur Rehman, Xiaoming Zhang, Faheem K. Butte, Shuai Feng, Bakhtiar Ul Haq, Buwen Cheng, Chuanbo Li. “Vanadium Based Zinc Spinel Oxides: Potential Material as Photoanode for Water Oxidation and Optoelectronic Devices.” **International Journal of Hydrogen Energy** 46, 55 (2021), 28110-28120.
30. Sajid Ur Rehman, Faheem K. Butt, Zeeshan Tariq, Xiaoming Zhang, Jun Zheng, Genadi Naydenov, Bakhtiar Ul Haq, and Chuanbo Li. "Elucidating the role of lattice thermal conductivity in π -phases of IV-VI monochalcogenides for highly efficient thermoelectric performance." **International Journal of Energy Research** 45, no. 4 (2021): 6369-6382.
31. Zeeshan Tariq, Sajid Ur Rehman, Junying Zhang, Faheem K. Butt, Xiaoming Zhang, Buwen Cheng, Sarwat Zahra, and Chuanbo Li. "Hierarchical mesoporous nanoflowers of Zn_2VO_4 for high capacity anode in lithium ion batteries." **Materials Science in Semiconductor Processing** 123 (2021): 105549.
32. Zeeshan Tariq, Sajid Ur Rehman, Xiaoming Zhang, Faheem K. Butt, Shuai Feng, Bakhtiar Ul Haq, Jun Zheng, Buwen Cheng, and Chuanbo Li. "Pristine and Janus monolayers of vanadium dichalcogenides: potential materials for overall water splitting and solar energy conversion." **Journal of Materials Science** 56, no. 21 (2021): 12270-12284.
33. Zia Ur Rehman, Faheem K. Butt, Narmina O. Balayeva, Faryal Idrees, Jianhua Hou, Zeeshan Tariq, Sajid Ur Rehman, Bakhtiar Ul Haq, Salem Alfaify, Saif Ali, Sher Zaman. "Two dimensional graphitic carbon nitride Nanosheets as prospective material for photocatalytic degradation of nitrogen oxides." **Diamond and Related Materials** 120 (2021): 108650.
34. Obeid, Mohammed M., Asadollah Bafekry, Sajid Ur Rehman, and Chuong V. Nguyen. "A type-II GaSe/HfS₂ van der Waals heterostructure as promising photocatalyst with high carrier mobility." **Applied Surface Science** (2020): 147607.
35. Bakhtiar Ul Haq, S. Alfaify, Thamraa Alshahrani, R. Ahmed, Faheem K. Butt, Sajid Ur Rehman, Zeeshan Tariq. “Devising square- and hexagonal-shaped monolayers of ZnO for nanoscale electronic and optoelectronic applications.” **Solar Energy** 211 (2020): 920-927.
36. Bakhtiar Ul Haq, S. Alfaify, A. Laref, R. Ahmed, Faheem K. Butt, Aijaz Rasool Chaudhry, Sajid Ur Rehman, and Q. Mahmood. "Optoelectronic properties of new direct bandgap polymorphs of single-layered Germanium sulfide." **Ceramics International** 45, no. 14 (2019): 18073-18078.
37. Sajid Ur Rehman, Faheem K. Butt, Zeeshan Tariq, Bakhtiar Ul Haq, Guochen Lin, and Chuanbo Li. "Cubic Germanium monochalcogenides (π -GeS and π -GeSe): emerging materials for optoelectronic and energy harvesting devices." **Solar Energy** 185 (2019): 211-221.

38. Zeeshan Tariq, Faheem K. Butt, **Sajid Ur Rehman**, Bakhtiar Ul Haq, F. Aleem, and Chuanbo Li. "First-principles study of electronic and optical properties of sulfur doped tin monoxide: A potential applicant for optoelectronic devices." **Ceramics International** 45, no. 6 (2019): 7495-7503.
39. Siyu Cao, Yue Zhao, **Sajid Ur Rehman**, Shuai Feng, Yuhua Zuo, Chuanbo Li, Lichun Zhang, Buwen Cheng, and Qiming Wang. "Theoretical studies on InGaAs/InAlAs SAGCM avalanche photodiodes." **Nanoscale research letters** 13, no. 1 (2018): 1-15.
40. Rabilah Gilani, **Sajid Ur Rehman**, Faheem K. Butt, Bakhtiar Ul Haq, and F. Aleem. "Elucidating the first-principles calculations of SnO₂ Within DFT framework and beyond: a library for optimization of various pseudopotentials." **Silicon** 10, no. 5 (2018): 2317-2328.
41. **Sajid Ur Rehman**, Faheem K. Butt, Bakhtiar Ul Haq, Salem AlFaify, Waheed S. Khan, and Chuanbo Li. "Exploring novel phase of tin sulfide for photon/energy harvesting materials." **Solar Energy** 169 x` (2018): 648-657.
42. **Sajid Ur Rehman**, Faheem K. Butt, Chuanbo Li, Bakhtiar Ul Haq, Zeeshan Tariq, and F. Aleem. "First-principles calculations of nitrogen-doped antimony triselenide: A prospective material for solar cells and infrared optoelectronic devices." **Frontiers of Physics** 13, no. 3 (2018): 1-12.
43. **Sajid Ur Rehman**, Faheem K. Butt, Fateh Hayat, Bakhtiar Ul Haq, Zeeshan Tariq, F. Aleem, and Chuanbo Li. "An insight into a novel cubic phase SnSe for prospective applications in optoelectronics and clean energy devices." **Journal of Alloys and Compounds** 733 (2018): 22-32.
44. Faheem K. Butt, Bakhtiar Ul Haq, **Sajid Ur Rehman**, R. Ahmed, Chuanbao Cao, and S. AlFaifi. "Investigation of thermoelectric properties of novel cubic phase SnSe: a promising material for thermoelectric applications." **Journal of Alloys and Compounds** 715 (2017): 438-444.
45. **Sajid Ur Rehman**, Faheem K. Butt, Zeeshan Tariq, Fateh Hayat, Rabilah Gilani, and F. Aleem. "Pressure induced structural and optical properties of cubic phase SnSe: An investigation for the infrared/mid-infrared optoelectronic devices." **Journal of Alloys and Compounds** 695 (2017): 194-201.

(b) Cover Pages

- **Sajid Ur Rehman**, Faheem K. Butt, Zeeshan Tariq, Xiaoming Zhang, Jun Zheng, Genadi Naydenov, Bakhtiar Ul Haq, Chuanbo Li "Elucidating the role of lattice thermal conductivity in π -phases of IV-VI monochalcogenides for highly efficient thermoelectric performance" **International Journal of Energy Research** (2020).

(c) Books/Book Chapters

1. Zeeshan Tariq, **Sajid Ur Rehman**, Xiaoming Zhang, Chuanbo Li, “Metal Oxide-carbon Composites of Vanadium Oxide and their Energy Storage Applications: Metal Oxide-Carbon Hybrid Materials”, **Elsevier Publisher**: Book Chapter (2022).
2. Saif Ali, Faheem K. Butt, Junaid Ahmad, Zia Ur Rehman, Sami Ullah, Mashal Firdous, **Sajid Ur Rehman**, Zeeshan Tariq “Recent Developments in Group II-VI Based Chalcogenides and Their Potential Application in Solar Cells.” 2D Nanomaterials Chemistry and Properties, **CRC (2022)**. (Book Chapter)
3. Faheem K. Butt, Sami Ullah, Junaid Ahmad, **Sajid Ur Rehman**, and Zeeshan Tariq. "Graphitic Carbon Nitride/Metal Oxides Nanocomposites and Their Applications in Engineering." In Composite Materials: Applications in Engineering, Biomedicine and Food Science, pp. 231-265. **Springer**, Cham, 2020.
4. Faheem K. Butt, **Sajid Ur Rehman**. Zinc Based Spinel Oxides for Energy Conversion and Storage Applications. *Nanotechnology: Applications in Energy, Drug and Food*, 31-48. **Springer**, Book Chapter (2019)
5. **Sajid Ur Rehman**., Butt, F. K., Tariq, Z., Li, C., & Li, C.. Tin-Based Novel Cubic Chalcogenides: A New Paradigm for Photovoltaic Research. *Emerging Photovoltaic Materials: Silicon & Beyond*, 141-163. **Wiley Publisher**: Book Chapter (2018)

7. References

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